

STATEMENT OF PURPOSE – AMAZON ML SUMMER SCHOOL

Before college, I had often heard about AI and ML, and naturally they caught my attention. I understood what they were supposed to do, but one question stayed with me: how? I watched videos, read articles, searched endlessly, and even asked AI about AI before I slowly began understanding what was happening beneath all the hype. My first structured exposure came through a 14-day Python and AI/ML workshop organised in my third semester, where I realised that being familiar with the terminology was very different from actually understanding and applying it.

Wanting to learn more, I participated in a few hackathons where, with the help of my teammates and my own curiosity, I gained practical exposure to machine learning. In one of them, we built a model to analyse multilingual sentiment on Twitter. Since India has many regional languages, we first translated the text into English using the Google Translate API. After cleaning and preprocessing the data, we performed sentiment analysis using Hugging Face's RoBERTa model along with predefined keywords. Working on this project taught me that, despite having powerful pre-trained models, fitting them effectively into real-world problems is not always straightforward. For example, our model struggled to distinguish sarcastic comments from genuine ones. Even with its limitations, building and experimenting with the project taught me far more than simply recreating something from a tutorial.

Despite these experiences, which have strengthened my interest in the field, I still hesitate to confidently say, "Yes, I know ML." I have enough exposure to appreciate what the field can do, but I also know how much I still do not understand. Rather than letting that discourage me, I take it as motivation to learn more. I want to build a stronger understanding of the principles behind how models learn, how they are trained and evaluated, and how to choose appropriate approaches for different problems rather than relying solely on existing tools.

I believe the Amazon ML Summer School will help me bridge that gap. Learning from and interacting with scientists at Amazon, who have extensive experience in the field, will provide me with the structured guidance I have been seeking. More importantly, this intensive program will help me strengthen the foundations that I currently lack while building on the curiosity and initiative that have brought me this far. Although my experience in ML is still developing, I believe my willingness to seek opportunities, learn through experimentation, and reflect on what I do and do not know makes me a strong fit for this program. After the program, I hope to apply these learnings to future projects, explore machine learning more deeply, be able to create my own trained models and eventually turn this interest into the field I pursue in the future.

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